

Chapter 5

Flow Matching and Similar Techniques

Exam Study Guide, Worked Concepts, Sample Questions, and

Prepared from	uploaded course materials
Main source	Chapter_5_Flow_Matching.ipynb
Exam source	Chapter_5_Summary_and_Sample_Questions.pdf
Practice source	Chapter_5_Flashcards.pdf

How to use it

Study this chapter as probability transport. Do not treat flow matching as only regression; the regression trains a velocity field whose ODE transports samples from base noise to data.

Study order: First read the concept sections, then solve the five sample exam questions, then test yourself with the flashcards.

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- 1. Quick exam map
- 2. Flow matching as transport
- 3. Velocity fields and ODE sampling
- 4. Bridge paths and target velocities
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Most useful mental shortcut
Diffusion learns denoising/score over a stochastic process; flow matching learns a velocity field for deterministic ODE transport.

1. Quick exam map

Flow matching treats generation as moving probability mass from a simple base distribution to a complex data distribution.

Base sample x_0

Gaussian noise or

simple distribution

Data sample x_1

target data example

Bridge x_t

intermediate state between x_0 and x_1 Velocity v_θ direction and speed to move

ODE solve

transport sample into data Figure: the main method pipeline for this chapter.

Exam topic	What you must be able to say
Velocity field	$v_\theta(x,t)$ tells each particle where to move and how fast.
ODE sampling	$dx/dt=v_\theta(x,t)$ transports noise to data.
Bridge/path	x_t defines intermediate states between start and endpoint.
Continuity equation	Describes how the whole density moves under the velocity field.

Exam topic	What you must be able to say
Latent flow	Learn transport in autoencoder latent space rather than pixel space.
OT couplings	Pair base and data samples coherently to simplify trajectories.

2. Zero-prior explanation: understand the

idea before the equations This section assumes you have not read the lecture yet. The goal is to build intuition first, then connect the intuition to the equations and exam language.

Flow matching as transport

The model does not directly output an image. It learns a time-dependent velocity field. A sample behaves like a particle carried by a wind field from simple noise toward data.

Velocity fields and ODEs

The learned field $v_\theta(x,t)$ defines $dx/dt=v_\theta(x,t)$. Sampling means numerically solving this ODE starting from x_0 sampled from the base distribution.

Bridge and target velocity

Training uses synthetic paths between start and end points. For a linear bridge, $x_t=(1-t)x_0+t x_1$ and the target velocity is x_1-x_0 .

Memory line

Flow matching learns the wind field that carries noise into data.

3. Core concepts and worked explanations

Continuity equation

The continuity equation describes how the entire probability density moves under the velocity field. This is why flow matching is distribution transport, not just pointwise regression.

Latent flow matching

Instead of learning in pixel space, learn the flow in an autoencoder latent space. This can reduce dimensionality and focus the transport on semantic structure.

OT and consistency ideas

Optimal-transport couplings choose better pairings between base and data samples, often making paths shorter and smoother. Consistency models aim for very few-step generation by learning stable maps across times.

4. Method explanation, worked example,

and application

Step	Method explanation
Define endpoints	Pair a base sample x_0 with a data sample x_1 .
Construct bridge	Create x_t between x_0 and x_1 .
Train velocity	Regress $v_\theta(x_t, t)$ toward known target velocity.
Sample by ODE	Initialize from p_0 and integrate to $t=1$.

Worked example

Example: in a two-moons dataset, random Gaussian points follow learned arrows toward the two-moons shape. The network is learning how the whole cloud should move, not simply denoising each point.

Applications and usage

Usage: flow matching is used for fast generative modeling, latent generative models, ODE-based sampling, conditional generation, and transport-based alternatives to diffusion.

5. Formula / definition cheat sheet

Read formulas as compact sentences. Do not memorize symbols blindly; use the symbol key table after the formula table to understand what each part represents.

Formula	Interpretation
$dx/dt=v_\theta(x,t)$ ODE defining sample motion.	$x_t=(1-t)x_0+t x_1$ Linear bridge.
$u_t=x_1-x_0$ Target velocity for linear bridge.	$L=E[v_\theta(x_t,t)-u_t ^2]$ Flow matching loss.
$\partial_t p_t + \nabla \cdot (p_t v_t)=0$ Continuity equation.	$x_{k+1}=x_k+\Delta t v_\theta(x_k, t_k)$
Euler update.	$v_{\text{guided}}=v_{\text{uncond}}+w(v_{\text{cond}}-v_{\text{uncond}})$ CFG for velocity fields.

6. Equation symbol key table

This table explains the main symbols used in the equations. When studying, cover the meaning column and check whether you can explain each symbol clearly.

Symbol	Meaning	Plain-English reading
x_0	Base sample	Starting noise/sample from simple
distribution.	x_1	Data endpoint
Target sample from data distribution.	x_t	Intermediate point
Position on the path at time t .	t	Continuous time
Progress from base at 0 to data at 1.	v_θ	Learned velocity

Symbol	Meaning	Plain-English reading
Network prediction of direction and	speed.	ut
Target velocity	Supervised training target.	pt
Density at time t	Distribution of particles at time t.	Δt
Solver step size	Small time increment for numerical	integration.

7. Main code prompt / implementation

skeleton This is not meant to replace the lecture code. It is a compact prompt showing the computational pattern you should recognize in homework and exams.

Flow matching training prompt

```
x0 = sample_base(batch_size) x1 = sample_data(batch_size) t = uniform(0, 1) xt = (1 - t) * x0 + t * x1
u_target = x1 - x0 v_pred = model(xt, t)
loss = mse(v_pred, u_target)
```

How to explain the code in words

Start by identifying the input, the model output, the objective being minimized, and the update rule. Then connect the code back to the formulas and the method table.

8. Common exam traps

Trap	Correct way to think
Calling flow matching just	regression Regression trains a field that induces distribution transport.
Ignoring bridge choice	The bridge changes intermediate states and target velocities.
Confusing score and velocity	Diffusion often estimates score/noise; flow matching estimates velocity.

9. Exam-style sample questions with

answers These are written in the style of conceptual exam answers. Use them to practice explaining, not only recognizing terms.

Question 1

Why is flow matching a transport model?

Sample answer

The learned velocity field moves an entire distribution from base to data.

Question 2

Why is the bridge not trivial?

Sample answer

It determines intermediate states and target velocities, changing learning difficulty.

Question 3

Why can latent flow be preferable?

Sample answer

Latent space can be lower-dimensional and semantically cleaner than pixel space.

Question 4

Why do OT couplings matter?

Sample answer

They pair samples more coherently, reducing tangled trajectories.

Question 5

How is CFG different in flow matching?

Sample answer

It guides a velocity field rather than a noise/score prediction.

10. Flashcards for rapid review

Use these for active recall. Cover the answer column and try to answer before checking.

Card 1

Q: What does $v\theta(x,t)$ represent?

A: Direction and speed of movement.

Card 2

Q: What is ODE sampling?

A: Solving $dx/dt=v\theta(x,t)$.

Card 3

Q: What is a bridge?

A: Path connecting base and data samples.

Card 4

Q: What does continuity equation express?

A: Conservation of probability mass.

Card 5

Q: What is latent flow matching?

A: Flow in latent space.

Card 6

Q: What does OT improve?

A: Pairing geometry.

Card 7

Q: What is Euler sampling?

A: Stepwise numerical ODE approximation.

Card 8

Q: What is the linear target velocity?

A: $x_1 - x_0$.

Card 9

Q: What must you remember about Velocity field?

A: $v_\theta(x, t)$ tells each particle where to move and how fast.

Card 10

Q: What must you remember about ODE sampling?

A: $dx/dt = v_\theta(x, t)$ transports noise to data.

Card 11

Q: What must you remember about Bridge/path?

A: x_t defines intermediate states between start and endpoint.

Card 12

Q: What must you remember about Continuity equation?

A: Describes how the whole density moves under the velocity field.

Card 13

Q: What must you remember about Latent flow?

A: Learn transport in autoencoder latent space rather than pixel space.

Card 14

Q: What must you remember about OT couplings?

A: Pair base and data samples coherently to simplify trajectories.

Card 15

Q: What does this formula represent:

$dx/dt = v_\theta(x, t)$?

A: ODE defining sample motion.

Card 16

Q: What does this formula represent:

$x_t = (1-t)x_0 + t x_1$?

A: Linear bridge.

Card 17

Q: What does this formula represent: $u_t = x_1 - x_0$?

A: Target velocity for linear bridge.

Card 18

Q: What does this formula represent:

$$L = E[||v\theta(xt,t) - ut||^2]?$$

A: Flow matching loss.

Card 19

Q: What does this formula represent: $\partial_t \rho t + \nabla \cdot (\rho t v_t) = 0$?

A: Continuity equation.

Card 20

Q: What does this formula represent:

$$x_{k+1} = x_k + \Delta t v\theta(x_k, t_k)?$$

A: Euler update.

Card 21

Q: What does this formula represent:

$$v_{\text{guided}} = v_{\text{uncond}} + w(v_{\text{cond}} - v_{\text{uncond}})?$$

A: CFG for velocity fields.

Card 22

Q: What does x_0 mean?

A: Base sample. Starting noise/sample from simple distribution.

Card 23

Q: What does x_1 mean?

A: Data endpoint. Target sample from data distribution.

Card 24

Q: What does x_t mean?

A: Intermediate point. Position on the path at time t .

Card 25

Q: What does t mean?

A: Continuous time. Progress from base at 0 to data at 1.

Card 26

Q: What does $v\theta$ mean?

A: Learned velocity. Network prediction of direction and speed.

Card 27

Q: What does u_t mean?

A: Target velocity. Supervised training target.

Card 28

Q: What does ρ_t mean?

A: Density at time t . Distribution of particles at time t .

Card 29

Q: What does Δt mean?

A: Solver step size. Small time increment for numerical integration.

Card 30

Q: What is the key idea of Flow matching as transport?

A: The model does not directly output an image.

It learns a time-dependent velocity field. A sample behaves like a particle carried by a wind field from simple noise toward

Card 31

Q: What is the key idea of Velocity fields and ODEs?

A: The learned field $v_\theta(x,t)$ defines $dx/dt=v_\theta(x,t)$.

Sampling means numerically solving this ODE starting from x_0 sampled from the base

Card 32

Q: What is the key idea of Bridge and target velocity?

A: Training uses synthetic paths between start and end points. For a linear bridge, $x_t=(1-t)x_0+t x_1$ and the target velocity is

Card 33

Q: What is the key idea of Continuity equation?

A: The continuity equation describes how the entire probability density moves under the velocity field. This is why flow matching is distribution transport, not just pointwise

Card 34

Q: What is the key idea of Latent flow matching?

A: Instead of learning in pixel space, learn the flow in an autoencoder latent space. This can reduce dimensionality and focus the transport on semantic

Card 35

Q: What is the key idea of OT and consistency ideas?

A: Optimal-transport couplings choose better pairings between base and data samples, often making paths shorter and smoother. Consistency models aim for very few-step generation by learning stable maps across

Card 36

Q: What happens in the step: Define endpoints?

A: Pair a base sample x_0 with a data sample x_1 .

Card 37

Q: What happens in the step: Construct bridge?

A: Create x_t between x_0 and x_1 .

Card 38

Q: What happens in the step: Train velocity?

A: Regress $v_\theta(x_t,t)$ toward known target velocity.

Card 39

Q: What happens in the step: Sample by ODE?

A: Initialize from p_0 and integrate to $t=1$.

Card 40

Q: Can you explain flow vs diffusion?

A: Velocity ODE vs denoising stochastic process.

Card 41

Q: Can you write a linear bridge?

A: $x_t=(1-t)x_0+t x_1$.

Card 42

Q: Can you explain sampling?

A: Initialize noise and integrate velocity.

Card 43

Q: Can you explain why bridge matters?

A: It defines supervision geometry.

Card 44

Q: Why is flow matching a transport model?

A: The learned velocity field moves an entire distribution from base to data.

Card 45

Q: Why is the bridge not trivial?

A: It determines intermediate states and target velocities, changing learning difficulty.

Card 46

Q: Why can latent flow be preferable?

A: Latent space can be lower-dimensional and semantically cleaner than pixel space.

Card 47

Q: Why do OT couplings matter?

A: They pair samples more coherently, reducing tangled trajectories.

Card 48

Q: How is CFG different in flow matching?

A: It guides a velocity field rather than a noise/score prediction.

Card 49

Q: In plain English, why is $dx/dt=v\theta(x,t)$ important?

A: It captures: ODE defining sample motion. Use it as a compact exam sentence, then define each symbol.

Card 50

Q: In plain English, why is $x_t=(1-t)x_0+t x_1$ important?

A: It captures: Linear bridge. Use it as a compact exam sentence, then define each symbol.

Card 51

Q: In plain English, why is $u_t=x_1-x_0$ important?

A: It captures: Target velocity for linear bridge.

Use it as a compact exam sentence, then define each symbol.

Card 52

Q: In plain English, why is $L=E[||v\theta(x_t,t)-u_t||^2]$ important?

A: It captures: Flow matching loss. Use it as a compact exam sentence, then define each symbol.

Card 53

Q: In plain English, why is $\partial_t \rho_t + \nabla \cdot (\rho_t v_t)=0$ important?

A: It captures: Continuity equation. Use it as a compact exam sentence, then define each symbol.

Card 54

Q: In plain English, why is $x_{k+1}=x_k+\Delta t v\theta(x_k,t_k)$ important?

A: It captures: Euler update. Use it as a compact exam sentence, then define each symbol.

Card 55

Q: In plain English, why is $v_{\text{guided}}=v_{\text{uncond}}+w(v_{\text{cond}}-v_{\text{uncond}})$ important?

A: It captures: CFG for velocity fields. Use it as a compact exam sentence, then define each symbol.

Explanation

Chapter 5: Flow Matching and Similar Techniques

How this section is organized

Each item below is a heading taken from the corresponding Jupyter notebook in the same order it appears there. The explanation beside it tells you what that notebook section is trying to teach and why it matters for the review.

Source notebook

Chapter_5_Flow_Matching.ipynb

Headings included

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Notebook headings with explanation

1. Chapter 5 - Flow Matching and Similar Techniques	After this chapter you should be able to: 1. Understand that the general idea and intuition behind flow matching 2.
2. Introduction to Flow Matching	The Basic Idea Flow matching learns a time-dependent velocity field $v_{\theta}(x,t)$ that tells you how to move particles from a simple base distribution (noise) to a complex target distribution (data). - Start distribution (easy to sample): $x_0 \sim p_0(x)$ (e.g., $N(0,I)$). - Target distribution (data): $x_1 \sim p_1(x)$ (unknown density, but we have samples). If we can learn a good velocity field, then we can sample from the data by: 1) sample $x_0 \sim p_0$ 2) integrate an ODE forward in time: $dx/dt =$.
3. The continuity equation	The continuity equation is the mathematical statement of a simple idea: > Probability mass can move around, but it can't appear out of nowhere or vanish. It's the same conservation law used for fluids ("mass conservation"), except here the "fluid" is probability density.
4. v_{θ} is a vector	In flow matching, $v_{\theta}(x,t)$ is a velocity vector. For data points $x \in \mathbb{R}^d$, the model is a vector field: $v_{\theta}: \mathbb{R}^d \times [0,1] \rightarrow \mathbb{R}^d$; $v_{\theta}(x,t) = \begin{bmatrix} v_{\theta,1}(x,t) \\ \vdots \\ v_{\theta,d}(x,t) \end{bmatrix}$. - The direction of $v_{\theta}(x,t)$ tells the particle which way to move in space. - The magnitude (speed) is the vector norm: $ v_{\theta}(x,t) _2$, which tells how fast it moves.
5. Flow Matching Intuition	The wind metaphor Imagine a huge field full of tiny balloons (particles). At time $t=0$ the balloons are scattered according to an easy distribution p_0 (like "random noise").
6. How Flow Matching Avoids Score Calculation	The "score" bottleneck in diffusion-style methods In many diffusion models, the key object is the score function of the noisy distribution: $\nabla_x \log p_t(x)$, i.e., the gradient of the log-density at time t which is related to the noise. A common training objective (in one parameterization) learns a network that is equivalent to the score by predicting noise: $x_t = \alpha(t)x_0 + \sigma(t)\epsilon$, $\epsilon \sim N(0,I)$, and training $\epsilon_{\theta}(x_t,t) \approx \epsilon$.
7. Flow Matching in 1D — Normal Shift and Bimodal Demo	This notebook demonstrates Flow Matching in 1D with two targets: 1. Normal shift: $p_{\text{data}}(x) = N(\mu, 1)$ with $\mu=3$ 2.
8. What one should expect	In the Normal shift case, the learned velocity should become nearly constant (approx μ). - In the Bimodal case, the learned velocity becomes nonlinear in x and evolves with t , pushing mass toward -2 and $+2$. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.

9. Example: Normal to another Normal Distribution	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
10. Example: Single Normal to Trimodal Distribution	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
11. Flow Matching for MNIST: Design Rationale	This notebook heading introduces flow matching for mnist: design rationale and explains how it supports the chapter concepts, equations, method steps, or implementation workflow.
12. Goal	Train a model to generate MNIST digits by learning a time-dependent velocity field (a "wind field"): $dx/dt = v_{\theta}(x,t); t \in [0,1]$. If we start from noise $x(0) \sim N(0,I)$ and integrate the ODE to $t=1$, then $x(1)$ should look like an MNIST digit. The notebook also includes 3 nearby code cells for implementation, demonstration, or visualization of this heading.
13. Oxford Flowers II Example	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 8 nearby code cells for implementation, demonstration, or visualization of this heading.
14. Printing the various parameters of the trained model	This notebook heading introduces printing the various parameters of the trained model and explains how it supports the chapter concepts, equations, method steps, or implementation workflow. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
15. Latent Flow Matching (LFM)	Latent Flow Matching combines two ideas: 1. Learn a compact latent space where the data becomes easier to model (via an autoencoder / VAE).
16. Latent flow matching Mathematics	Define distributions in latent space Let $x \sim p_{\text{data}}$. - AE latents: $z_1 = E_{\psi}(x)$. - VAE latents: $z_1 \sim q_{\psi}(z_{\text{mid}} x)$. Choose a simple base distribution in latent space: $z_0 \sim p_0(z) = N(0,I)$.
17. Latent Flowmatching Example for Oxford Flowers II	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 4 nearby code cells for implementation, demonstration, or visualization of this heading.
18. Using VAE Instead	The code below shows how to do the same with a VAE. However, it is much more difficult to train the VAE to generate appropriate images. The notebook also includes 5 nearby code cells for implementation, demonstration, or visualization of this heading.
19. Class-Conditional, Unguided, and Classifier-Free Guided Generation in Flow Matching	There are three obvious ways: 1) Unguided (unconditional) flow matching 2) Class-conditional (unguided) generation 3) Classifier-free guidance (CFG) for *guided* generation We assume the standard ODE generative model: $dx/dt = v_{\theta}(x,t); t \in [0,1]$. Unguided / Unconditional Flow Matching Training Sample: - noise (source): $x_0 \sim p_0 = N(0,I)$ - data (target): $x_1 \sim p_{\text{data}}$ - time: $t \sim \text{Unif}(0,1)$ Define a straight-line path: $x_t = (1-t)x_0 + t x_1$.
20. Conditional Flow Matching Example (Flowers 102)	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 2 nearby code cells for implementation, demonstration, or visualization of this heading.
21. Classifier Guided Flow Matching Example (Flowers 102)	Training a small custom classifier Training a more powerful classifier using transfer learning The small classifier did not work well. We can use transfer learning to train a more competent classifier. The notebook also includes 3 nearby code cells for implementation, demonstration, or visualization of this heading.

22. Sampling with Classifier Guidance	This notebook heading introduces sampling with classifier guidance and explains how it supports the chapter concepts, equations, method steps, or implementation workflow. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
23. Classifier-Free Guidance Example (Flowers 102)	This heading introduces a practical example or code activity. Its purpose is to connect the theory to an executable workflow or visualization. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
24. Optimal-Transport (OT) Couplings for Flow Matching	You already know rectilinear (straight-line) flow matching: we pick endpoints $x_0 \sim p_0$ (noise) and $x_1 \sim p_{\text{data}}$ (data), define a straight path $x_t = (1-t)x_0 + tx_1$, and train a velocity field $v_\theta(x,t)$ to match the path derivative. OT-based flow matching keeps the *same* straight-line path but changes the pairing between noise samples and data samples.
25. Hungarian Algorithm	Problem it solves You have a cost matrix C in $R^{n \times n}$ where C_{ij} is the cost of pairing item i (e.g., noise sample x_0^i) with item j (e.g., data sample x_1^j). You want a one-to-one matching (a permutation π) that minimizes total cost: $\min_{\pi} \sum_{i=1}^n C_{i,\pi(i)}$.
26. Example: Hungarian Algorithm (minibatch OT for flow matching)	Assume we sampled a minibatch: - 2 noise points (source) $x_0^1, x_0^2 \sim N(0,1)$ - 2 data points (target) $x_1^1, x_1^2 \sim p_{\text{data}}$ We build a cost matrix using squared distance: $C_{ij} = x_0^i - x_1^j ^2$. In minibatch OT assignment, we choose the one-to-one matching (a permutation) minimizing: $\min_{\pi} \sum_{i=1}^2 C_{i,\pi(i)}$. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
27. Consistency Methods for Generation	Consistency methods are generative models that learn a mapping which is self-consistent across noise levels (or “times”), enabling very fast sampling—often 1–2 steps. They can be viewed as a bridge between: - diffusion / score models (many-step denoising), - rectified flows / ODE transport (continuous-time deterministic sampling), - and distillation (compressing many steps into a few).
28. Summary	Consistency methods enforce: - time-invariant denoising / canonicalization - by training $f_\theta(x,t)$ so that its output is consistent across noise levels: $f_\theta(x_{t_1}, t_1) \approx f_\theta(x_{t_2}, t_2)$. This enables one-step or few-step generation, making them a powerful acceleration path beyond standard flow matching or rectified flows.
29. Example: 1D Consistency Model on a Bimodal Distribution	We model a 1D bimodal distribution: $p_{\text{data}}(x) = t_1/2N(-\mu, \sigma_{\text{data}}^2) + t_1/2N(+\mu, \sigma_{\text{data}}^2)$. Forward noising (closed-form marginals): $x_t = \alpha(t)x_0 + \sigma(t)\epsilon$, $\epsilon \sim N(0,1)$, with a schedule satisfying: $\alpha(t)^2 + \sigma(t)^2 = 1$. The notebook also includes 4 nearby code cells for implementation, demonstration, or visualization of this heading.
30. Teacher-Student Consistency Models	Why do we need a teacher in the first place? A *pure* consistency objective tries to make a model output invariant across time: $f_\theta(x_s, s) \approx f_\theta(x_t, t)$; ($s < t$) for noisy states x_s, x_t that come from the same underlying clean sample x_0 .
31. Sampling (fast generation)	After training, sampling is fast because the student is trained to map noisy states to clean predictions directly. - One-step: $x_T \sim p_{\text{noise}}$; $\hat{x}_0 \sim p_\theta(x_0 x_T, T)$ or $\hat{x}_0 = f_\theta(x_T, T)$. - Few-step (optional): evaluate the student at intermediate times before T for better quality, then distill down to 1-step if desired.
32. Summary Comparision Table	> Notation (generic): > - Data: $x \in R^D$, latent: $z \in R^d$ > - Time: $t \in [0,1]$ (or $1, \dots, T$) > - Score: $s_t(x) = -\log p_t(x)$ > - Velocity (wind): $v_t(x)$ > - Condition (prompt/label/etc.): c Family / Variant; What the NN estimates (primary object); Sampling mechanism; “Hard parameter” / what’s difficult; Typical tradeoffs (pros / cons) VAE (standard); Encoder posterior $q_\phi(z x)$ (often $\mu_\phi(x), \log \sigma_\phi^2(x)$) and decoder likelihood $p_\theta(x z)$; Sample $z \sim p(z)$ then decode $x \sim p_\theta(x z)$.

33. What each method is “trying to estimate” (one-line view)	VAE: learns an <i>approximate posterior</i> $q_{\phi}(z x)$ and a <i>decoder likelihood</i> $p_{\theta}(x z)$. - Diffusion: learns a <i>denoiser</i> (often $\text{var}_{\epsilon}\epsilon_{\theta}$) that is equivalent to a <i>score-like</i> object $\nabla_x \log p_t(x)$. - Flow Matching: learns a <i>velocity / wind</i> field $v_{\theta}(x,t)$ directly (no explicit score). - Consistency: learns a <i>canonical map</i> $f_{\theta}(x_t, t)$ that should be invariant across time; teacher-free risks collapse, teacher-based anchors the target.
34. Where “difficulty” typically lives	VAE: posterior approximation + likelihood mismatch (ELBO gap), potential posterior collapse. - Diffusion: score-like estimation across many noise levels + expensive sampling steps; guidance needs careful tuning. - Flow Matching: picking a good bridge/coupling and learning a globally consistent vector field; ODE solver accuracy. - Consistency: identifiability/collapse without a teacher; with teacher the main difficulty shifts to distillation and teacher quality.
35. Further Readings	Category; What the neural network estimates; Key paper VAE; Encoder posterior parameters $q_{\phi}(z x)$ (e.g., $\mu_{\phi}(x), \log \sigma_{\phi}^2(x)$) and decoder likelihood $p_{\theta}(x z)$; Kingma & Welling (2013) — <i>Auto-Encoding Variational Bayes</i> (https://arxiv.org/abs/1312.6114) Diffusion (Unguided); Typically noise predictor $\text{var}_{\epsilon}\epsilon_{\theta}(x_t, t)$ (equivalent to learning a score-like object); Ho, Jain & Abbeel (2020) — <i>Denoising Diffusion Probabilistic Models</i> (https://arxiv.org/abs/2006.11239) Diffusion.
36. Flow Matching Variants	A useful way to organize most flow-matching / rectified-flow / stochastic-interpolant style methods is by which lever they change. Lever 0: the basic formulation Start distribution: $x_0 \sim p_0$ Target distribution: $x_1 \sim p_1$ Pick time: $t \sim \text{Uniform}[0,1]$ Baseline bridge (linear interpolant): $x_t = (1-t)x_0 + tx_1$ Baseline target velocity: $u_t = \frac{d}{dt}x_t = x_1 - x_0$ Flow matching loss: $L(\theta) = \mathbb{E}[\ v_{\theta}(x_t, t) - u_t\ _2^2]$.
37. When reading a new paper	When you encounter a “new” flow matching method, ask: 1. Coupling: Did they change how x_0 pairs with x_1? (OT/minibatch OT/learned pairing) 2.